When Is Periodic Detection Enough for UAV Traffic Monitoring? Theory and a Pre-Registered Test of Cue-Triggered Gating

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***Abstract*—Onboard detectors on traffic-monitoring UAVs can run on every S-th frame or only when a cheap cue signals a change. We ask when such a cue-triggered gate beats the oblivious periodic schedule at equal cost for newly appearing vehicles. A closed theory gives exact miss and onset-latency expressions for both policies, a gain–cost identity whose win set is an interval in the cue's sensitivity, an iso-KPI condition and a density and ego-motion impossibility result; together they confine a gate win to tight latency targets, sparse traffic, a hovering camera and a cheap, onset-selective cue. We tested this in a pre-registered experiment on UAVDT: parameters were estimated on the training split and the predicted signs of 72 cells frozen before the test split was opened. The frozen criteria are not violated (62 correct, 5 wrong, 5 inconclusive; 88.4% of decided cells correct without the 24 infeasible iso-KPI cells), but the predicted corner did not appear, no gate beat the periodic schedule at matched cost, and one refutation criterion could not be evaluated. Post hoc analyses trace this to weak cues, low recall at onset and an optimistic model. Design rule: measure the latency-miss floor, choose the periodic stride from the iso-miss requirement, and consider a gate only if its onset separation exceeds a threshold implied by the gain–cost identity.**

***Index Terms*—UAV traffic monitoring, onboard object detection, frame skipping, event-triggered sensing, pre-registered evaluation.**

I. Introduction

UNMANNED aerial vehicles (UAVs) are now an ordinary sensing layer of intelligent transportation systems: a hovering or cruising drone watches an intersection, a corridor or an incident scene and turns video into counts, trajectories and alerts [1]–[3]. The expensive step on board is the object detector. Its accuracy has improved steadily for small aerial objects [4]–[6], but its run rate, not its architecture, sets most of the compute and energy bill of an onboard pipeline [7], [8]. A practical question therefore precedes the choice of detector: on which frames should it run?

Two families of answers exist. An *oblivious periodic* schedule runs the detector on every S-th frame, whatever the scene does. An *adaptive gate* runs it only when a cheap cue, such as frame differencing, a border-entry cue or a tiny detector, signals that something may have changed [9]–[12]. Gating is attractive because most frames of a traffic video are uneventful. It is also the default intuition of the video-analytics literature, although careful studies on static cameras have already found that plain sampling with a full detector can beat proxy filters [13]. Aerial video makes the case for gating weaker still: the camera moves, so frame differences fire on the background [14], [15], and the scene is dense, so a cue that reacts to motion fires on vehicles that are already being tracked [16], [17].

This paper asks when, at equal cost, an oblivious periodic schedule is beaten by a cue-triggered gate for the detection of *newly appearing* vehicles from a UAV. New vehicles are the hard case for a periodic schedule: a vehicle that enters the footprint is invisible to the system until the next detector run, so the onset latency, not the overall recall, is the relevant key performance indicator (KPI). If a gate can win anywhere, it is here.

We answer the question in two steps. First, a short theory (Sections III–IV) shows that the cue sensitivities for which a gate wins form an interval that may be empty (Theorem 2, Corollary 2), sets the cost a gate must undercut at a target latency-miss (Corollary 4) and closes the interval as density and ego-motion grow (Theorem 3): a frame-level gate can win only with tight latency targets, sparse traffic, a hovering camera and a cheap, onset-selective cue. Second, we test that prediction in a pre-registered experiment on UAVDT [16]: all channel parameters were estimated on the training split, and the predicted sign of the gate advantage was frozen for 72 cells, with a three-way scoring rule and the refutation criteria, before the test split was opened (tag `prereg-28-v1`, 1 October 2026).

The pre-registered verdict is "confirmed" in the sense fixed in advance: 62 cells are correct, 5 wrong and 5 inconclusive (92.5% of the decided cells correct), and no evaluable refutation criterion holds. Two qualifications matter as much: the 24 iso-KPI cells are correct for an uninformative reason, because neither policy can reach the frozen target of 2% latency-miss on the test split (without them the share is 88.4%), and the density-boundary criterion could not be evaluated. The corner in which the model predicted a gate win did not appear, and no gate beat the periodic schedule with a confidence interval excluding zero in any of the 24 matched-cost cells; post hoc analyses trace this to weak cues, low recall at onset and a shift of the cue channel.

The paper makes three contributions.

Manuscript received —; revised —. (Corresponding author: Dat Lam Quoc.)

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- *Theory.* A closed theory of periodic versus gated detection for object-level onset events in which traffic density, ego-motion and detector recall enter explicitly: a kinematic certificate for the detector rate (Theorem 1), the gain–cost identity and its win-set interval, an iso-KPI condition and a density and ego-motion impossibility result (Sections III–IV).
- *Pre-registered test.* To our knowledge, the first pre-registered, out-of-sample test of gating on aerial video (literature search protocol in the repository): the family of cells was frozen before the test split was opened and scored by a three-way rule with a power criterion, and all deviations are reported (Sections V–VI, Appendices A–B).
- *Negative result with a mechanism.* Post hoc measurements quantify why the gate lost (weak cues, recall at onset, calibration of the model; Section VI-F) and yield a design rule for the detector rate (Section VII-A).

A separate paper by the author on frame-skipping latency certificates for edge devices is in press; every statement here is proved and tested independently.

II. Related Work

A. Frame Gating and Aerial Vehicle Detection

Video-analytics systems save inference by filtering frames with low-level differences [9], adapting resolution and frame rate [18], learning input filters or skipping policies [10], [11] or interleaving heavy and light detectors [19]; related schemes exist for tracking and low-power sensing [20], [21], and drones filter frames on board to save uplink bandwidth [12] or select keyframes for search and rescue [22]. Most of this work assumes a static camera and reports savings against processing every frame, not against a periodic schedule of equal cost. Seiden [13] is the exception: on static-camera queries, uniform sampling with the full model beats learned proxy filters. UAV traffic monitoring in this journal has concentrated on detector accuracy and tracking: reviews benchmark detectors and their latency [1], [2]; detectors target tiny objects [4], high flight altitude [5], lightweight inference [6], road regions [23] and cascades [24]; and tracking and trajectory pipelines have been evaluated in the field [3], [25]–[30]. Recent trackers compensate camera motion [14], [15] or run the detector at a fixed rate on an embedded board [31]; benchmarks document the density and camera motion of drone video [16], [17], [32]; latency-aware evaluation has been argued for in tracking and streaming perception [33]–[36]; and small, short-lived targets are common in aerial problems [37]–[39]. *Gap G1: to our knowledge, no work in this journal studies how often the onboard detector should run, or compares periodic and adaptive schedules at equal cost.*

B. Sampling and Scheduling Theory

Event-triggered sampling can outperform periodic sampling for first-order stochastic systems [40]; sensor activation under correlated events, optimal sampling for remote estimation and the age of incorrect information formalise when adaptive sampling pays [41]–[43], and adaptive sensing has fundamental limits relative to non-adaptive designs [44]. Our events are discrete objects with a kinematic dwell time, the adaptive policy is driven by an imperfect cue whose false-fire rate grows with traffic density, and the comparison is at equal total cost including the cue. *Gap G2: we are not aware of a pre-registered, out-of-sample test of a gating prediction on aerial video.*

III. Model and Certificates

This section states the look-count law of a periodic detector and the kinematic certificate. Proofs of all statements in Sections III–IV are in Supplementary S1.

A. Notation and Standing Assumptions

The capture stream has frame index $t \in \mathbb{Z}$ and capture rate F frames/s. An object-level event e has onset (birth) frame b_e and visible duration $D_e \geq 1$ frames. A periodic schedule with (real) stride $S \geq 1$ and phase $\varphi \sim U[0, S)$ runs the detector on frames $\lceil \varphi + kS \rceil$, $k \in \mathbb{Z}$; decimation by k followed by a detector every R retained frames is the periodic schedule with $S = kR$. N_e is the number of looks on the frames of e, r the per-look recall, a the activation (fraction of frames on which the detector runs; $a = 1/S$ for a periodic schedule), $f_{det} = aF = F/S$ the detector rate, L_e the onset latency (frames from b_e to the first successful look) and miss the probability that an event receives no successful look in its KPI window. The standing assumptions are: *A-ind* (independent looks): given the looked-at frames, each look on a frame of e succeeds independently with probability r; *A-phase*: the phase is uniform and independent of the events; *A-stat*: onsets and durations are independent of the observation window.

B. Look-Count Law and Censoring

Lemma 1 (Look-count law): Assume A-phase, a periodic schedule with stride $S \geq 1$ and an event of D consecutive frames. Write $D/S = n + f$ with $n = \lfloor D/S \rfloor$, $f \in [0, 1)$. Then (i) $N = n + \mathrm{Bernoulli}(f)$; (ii) under A-ind,

$$\mathrm{miss}_r(D, S) = \mathbb{E}[(1-r)^N] = (1-r)^n(1 - fr); \quad (1)$$

(iii) for $r = 1$, $\mathrm{miss}_1(D, S) = (1 - D/S)_+$.
Applied to the first $\min(D, L_{max} + 1)$ frames of an event, (1) gives the onset-latency law $P(L \leq L_{max}) = 1 - \mathrm{miss}_r(\min(D, L_{max} + 1), S)$ for integer S (Supplementary Corollary S1); $D \mapsto \mathrm{miss}_r(D, S)$ is non-increasing (Supplementary Lemma S1).

Corollary 1 (Right-censoring; birth window): Let a sequence have frames $1, \ldots, L_s$ and observe $D_{obs} = \min(D, L_s - b + 1)$. The birth window of width W keeps events with $1 < b \leq L_s - W$. (i) For $r = 1$ and $S \leq W$, $\mathrm{miss}_1(D_{obs}, S) = \mathrm{miss}_1(D, S)$ for every kept event. (ii) For $r < 1$, $\mathrm{miss}_r(D_{obs}, S) \geq \mathrm{miss}_r(D, S)$, with per-event bias at most $(1-r)^{\lfloor (W+1)/S \rfloor}$. (iii) Under A-stat the kept events are selected on onset time only (no length bias).

C. Kinematic Certificate (P1)

Assume (K1) flat ground and a nadir camera at height h with along-track field of view θ, footprint length $L(h) = 2h\tan(\theta/2)$; (K2) a point vehicle with constant speed $v \le v_{max}$ along a straight chord of length ℓ inside the footprint or a region of interest, in view for $\tau = \ell/v$; (K3) capture frames at times j/F and a detector on every S-th frame (integer S, arbitrary phase), $f_{det} = F/S$. Let $n(r,\varepsilon) = \lceil \ln\varepsilon / \ln(1-r) \rceil$ for $r < 1$ and $n = 1$ for $r = 1$.

Theorem 1 (P1, kinematic certificate): Under K1–K3 and A-ind, if

$$f_{det} \;\ge\; \frac{n(r,\varepsilon)\, v_{max}}{\ell}, \tag{2}$$

then for every phase and every vehicle with $v \le v_{max}$: $N \ge n(r,\varepsilon)$ and $\mathrm{miss} \le (1-r)^n \le \varepsilon$. For a full traverse, $\ell = L(h)$ and the condition reads $f_{det} \ge n\, v_{max}/(2h\tan(\theta/2))$.

The capture rate enters only through f_{det}: the requirement is on detector runs per second. Vehicles clipping a footprint corner admit no finite certificate, so it is stated for a region of interest crossed with chord $\ell \ge \ell_{min}$ (Supplementary Remark S1, which also covers an oblique camera). Altitude enters the certificate through $L(h)$ and the recall $r(h)$; an optimal altitude exists if recall at high altitude vanishes faster than $1/h$ (Supplementary Theorem S1).

IV. When Can a Gate Win?

This section derives when a frame-level gate can beat a periodic schedule of equal cost, and when density or ego-motion rules it out.

A. Gate Channel

Event e has onset window $O_e = [b_e, b_e + w)$; $I_t = 1$ if frame t lies in some O_e, and $\rho = \rho_{onset}(w)$ is the long-run fraction of frames with $I_t = 1$. *A-gate*: a frame-level cue fires $G_t \sim \mathrm{Bernoulli}(q_{in})$ if $I_t = 1$ and $\mathrm{Bernoulli}(q_{out})$ otherwise, independently across frames given I. *Refresh*: the detector also runs on frames $t \equiv \varphi \pmod R$, φ uniform ($R = \infty$: no refresh). *Policy* (gate $\vee$ refresh): run the detector iff $G_t = 1$ or t is a refresh frame; the cue costs c detector-equivalents per frame. *KPI window*: $T_e = \min(D_e, L_{max} + 1)$ for the latency KPI and $T_e = D_e$ for the miss KPI; its first $w' = \min(w, T_e)$ frames are the event's own onset window. *A-sparse*: the other $T_e - w'$ frames lie outside all onset windows. The periodic competitor at equal cost has activation $a_P = a_G + c$.

B. Activation, Miss and the Gain–Cost Identity

Proposition 1 (G1, activation): $a_G = \frac{1}{R} + \big(1 - \frac{1}{R}\big)\big[\rho\, q_{in} + (1-\rho)\, q_{out}\big]$.

Proposition 2 (G2, miss of gate $\vee$ refresh): Let $n_{in}(\varphi), n_{out}(\varphi)$ be the numbers of refresh frames among the first w' and the remaining $T_e - w'$ frames of the KPI window. Under A-gate, A-sparse and A-ind,

$$\mathrm{miss}_G(T_e) = \frac{1}{R}\sum_{\varphi=0}^{R-1} (1-r)^{n_{in}+n_{out}} \times (1 - rq_{in})^{w'-n_{in}} (1 - rq_{out})^{T_e - w' - n_{out}}. \tag{3}$$

For $r = 1$ and $T_e \le R$: $\mathrm{miss}_G = (1 - T_e/R)(1-q_{in})^{w'}(1 - q_{out})^{T_e - w'}$.

Remark 1 (G2′, multi-look advantage under imperfect detectors): Consider the latency KPI with window $T := T_e$, the gate of Proposition 2 with $R = \infty$ and fixed cue parameters, hence a fixed activation $a = a_G + c$, and the periodic competitor at the same activation with $Ta < 1$. Then $\Delta(r) = \mathrm{miss}_P(r) - \mathrm{miss}_G(r)$, with $\mathrm{miss}_P(r) = 1 - Ta\,r$ and $\mathrm{miss}_G(r) = (1 - rq_{in})^{w'}(1 - rq_{out})^{T-w'}$, is concave on $[0,1]$ with $\Delta(0) = 0$; hence there is $r^\dagger \in [0,1]$ such that the gate's advantage grows as the detector gets worse on $[r^\dagger, 1]$ and shrinks below $r^\dagger$. The periodic schedule places at most one look in the window; the gate places several consecutive looks, which are wasted when $r = 1$ but are extra chances when $r < 1$.

Theorem 2 (G3, gain–cost identity, $r = 1$): Let $T := T_e \le R$, $w' \ge 1$ and $\Delta := \mathrm{miss}_P - \mathrm{miss}_G$ with miss_P the periodic miss at activation $a_G + c$. If $T(a_G + c) \ge 1$ then $\Delta = -\mathrm{miss}_G \le 0$. Otherwise

$$\Delta = \underbrace{\Big(1 - \tfrac{T}{R}\Big)\Big[1 - (1-q_{in})^{w'}(1-q_{out})^{T-w'}\Big]}_{\text{gain } \mathcal{G}} - \underbrace{T\Big(1 - \tfrac{1}{R}\Big)\big[\rho q_{in} + (1-\rho)q_{out}\big]}_{\text{cost } \mathcal{C}} - Tc. \tag{4}$$

Corollary 2 (G4, threshold structure, $r = 1$): Fix $(q_{out}, \rho, R, w, T, c)$ with $T \le R$ and let $\kappa(R,T) = \frac{R-T}{T(R-1)}$ ($\kappa = 1/T$ for $R = \infty$). (i) The set of $q_{in} \in [0,1]$ for which the gate strictly wins is an interval (q_{in}^*, q_{in}^{up}), possibly empty and possibly bounded above. (ii) *Necessary condition* (N-G): $(1-\rho)\, q_{out} + \frac{c}{1 - 1/R} < \kappa(R,T)$. (iii) *Sufficient condition* (S-G): if $\rho + (1-\rho)q_{out} + \frac{c}{1-1/R} < \kappa(R,T)$, the gate with $q_{in} = 1$ wins. (iv) *First order* ($R = \infty$, small q): $\Delta \approx (q_{in} - q_{out})(w' - T\rho) - Tc$.

The first-order expression of Corollary 2(iv) is in fact an upper bound for every recall, which gives a screening condition that only involves measurable quantities; it is derived for this paper from Propositions 1–2 and Lemma 1.

Corollary 3 (Necessary onset separation): Let $R = \infty$, assume A-gate, A-sparse and A-ind, let $r \in (0,1]$, $T := T_e$, $w' = \min(w, T) > T\rho$ and $T(a_G + c) \le 1$, and let $J = q_{in} - q_{out}$. Then

$$\Delta \;\le\; r\big[J\,(w' - T\rho) - Tc\big], \tag{5}$$

so that $J \le J^* := Tc/(w' - T\rho)$ implies $\Delta \le 0$.

The threshold J^* does not depend on r: a worse detector scales both sides of the comparison equally to first order. The bound was checked numerically on random configurations (tests/test_p4.py); it is necessary, not sufficient. Corollary 3 was derived after the experiment for the design rule; it is not part of the pre-registration and is stated for $R = \infty$ (the experiment used $R = 30$), so it is a screening heuristic rather than a scored prediction.

C. Iso-KPI Comparison

At matched activation in the low-budget regime ($a < 1/T$) almost any informative cue wins to first order, because a

periodic schedule with $a < 1/T$ misses a fraction $1 - Ta$ of short windows; this is not the operating regime of an ITS deployment, where the budget is set by a target miss. The ITS-relevant comparison is iso-KPI.

Corollary 4 (G5, iso-KPI, ideal onset gate, $r = 1$): Let events have $D \geq T$, and let the onset gate have $w = 1$, $q_{in} = 1$, false-fire rate q_{out} and no refresh; $\rho_1 = \rho_{onset}(1)$. To reach latency-miss $\leq \varepsilon$ the periodic schedule needs $a_P = (1-\varepsilon)/T$, while the ideal onset gate reaches latency-miss zero at cost $\rho_1 + (1 - \rho_1)q_{out} + c$. The gate is cheaper at equal KPI iff

$$\rho_1 + (1 - \rho_1)\, q_{out} + c \;<\; \frac{1 - \varepsilon}{T}. \tag{6}$$

For $r < 1$ the onset gate must fire on at least $n(r, \varepsilon)$ window frames and the comparison is numerical.

D. Density and Ego-Motion Boundary (P3)

Lemma 2 (Effective false-fire rate): If, outside onset windows, the cue fires when the background/ego-motion residual fires (probability q_b) or when any of M other moving objects triggers it (probability q_o each), independently, then $q_{out,eff}(M) = 1 - (1 - q_b)(1 - q_o)^M$.

Theorem 3 (P3, density and ego-motion impossibility): Let $r = 1$ and $T \leq R$. (i) *Density.* With $q_{out}^{max} = \big(\kappa(R, T) - c/(1 - 1/R)\big)/(1 - \rho)$ and $q_b < q_{out}^{max} < 1$, for every

$$M \;\geq\; M_{nec} := \frac{\ln\big((1 - q_b)/(1 - q_{out}^{max})\big)}{-\ln(1 - q_o)} \tag{7}$$

no $q_{in} \in [0, 1]$ makes the gate beat the periodic schedule of equal cost; hence $M^* \leq \lceil M_{nec} \rceil$. If $q_b \geq q_{out}^{max}$, no gate wins at any M. (ii) *Ego-motion.* Without compensation $q_b \to 1$, so $q_{out,eff} \to 1$ and (N-G) fails for every M. With compensation at cost c per frame, (N-G) requires $(1 - \rho)q_{out,eff} + c/(1 - 1/R) < \kappa$, and for any budget $a < c + a_{min}$, $a_{min} = 1/R + (1 - 1/R)(1 - \rho)q_{out,eff}$, no gate configuration is feasible while a periodic schedule is. (iii) *Iso-KPI.* The ideal onset gate of Corollary 4 is cheaper at equal latency-miss iff $M < M_{iso} := \ln\big((1-q_b)/(1-Q)\big)/(-\ln(1-q_o))$ with $Q = \big((1 - \varepsilon)/T - \rho_1 - c\big)/(1 - \rho_1)$, and never if $Q \leq q_b$.
Since Δ need not be monotone in q_{out}, the exact M^* is computed numerically; M_{nec} is an upper bound.

V. Experimental Protocol

The protocol was fixed in a pre-registration whose hash and git tag were recorded before the test split was opened (Appendix A); Fig. 1 shows the pipeline.

A. Data and Splits

UAVDT [16] contains 50 sequences (30 train, 20 test), 40,421 frames at 30 frames/s and 2,653 vehicle tracks with occlusion and out-of-view flags. Every estimate that enters a prediction was made on the TRAIN split (30 sequences, 23,829 frames). Two data rules were fixed before freezing: frames beyond the last annotated frame of a sequence are discarded (they carry no labels, not zero vehicles), and the border band is computed from the true image width read from the images. On TEST all 20 sequences have annotations up to their last frame, so 0 frames were dropped (16,592 frames used). VisDrone2019-VID [17] is listed in the pre-registration as a second dataset and is left for a follow-up study; this paper is single-dataset (Section VII-C).

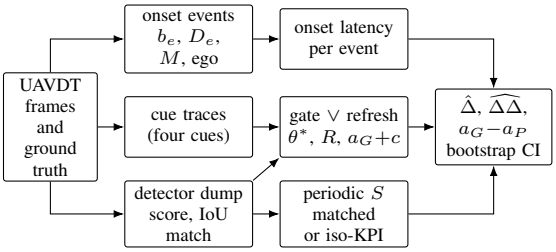


Fig. 1. Offline replay pipeline. Both policies are replayed on the same detector dump and the same cue traces; the gate pays the cue cost c on every frame and the periodic competitor is given the same total activation (matched cost) or the activation it needs to reach a latency-miss target (iso-KPI). All channel parameters and thresholds come from the TRAIN split; the TEST split is replayed once with the frozen values.

B. Onset Events, Density and Ego-Motion

An onset event (code name E3 in the repository and the pre-registration) is a new vehicle: a track that first becomes visible after the first frame of its sequence, by entering the image or by leaving occlusion. Its onset b_e is the first frame on which the box is visible (occlusion and out-of-view flags not set), and its duration D_e is the length of its first visible run. Events are kept if their onset leaves a birth window of 120 frames before the end of the sequence (Corollary 1). The density M is the number of visible vehicles at onset, binned into M 1–5, M 6–20 and $M > 20$. A sequence is *hovering* if the median image-centre shift of the frame-to-frame homography (ORB features, RANSAC) is below 0.61 px/frame and *moving* otherwise. The threshold is the equal-density point of a two-component Gaussian mixture fitted on the TRAIN sequences (modes 0.13 and 1.20 px/frame, Ashman $D = 2.92$); it gives 19 hovering and 11 moving TRAIN sequences and was applied unchanged to TEST.

C. Detectors

The main detector is YOLO26-s [45] trained on VisDrone2019-DET at input size 1024 (validation mAP_{50} 0.500), exported to OpenVINO [46] and run on the integrated GPU of an Intel Core i7 laptop. A detection is a box with score ≥ 0.25 (the library default), matched one-to-one to ground truth at IoU ≥ 0.5. The per-look recall r is the share of visible vehicle boxes that are matched, measured on TRAIN: 0.944 [0.913, 0.965] for YOLO26-s@1024. Hypothesis H8 needs a second, lower-recall detector: by a rule fixed before any recall was measured, a candidate level qualifies if its recall is at least 0.05 below the main level. Of YOLO26-n@1024, YOLO26-s@640 and YOLO26-n@640 (validation mAP_{50} 0.404 for YOLO26-n), only the last qualified, with $r = 0.876$ [0.829, 0.912].

D. Cues, Cost and Operating Points

Four frame-level cues were traced on every frame: *raw difference* (share of changed pixels between consecutive frames

TABLE I
FROZEN OPERATING POINTS (UAVDT TRAIN, 30 SEQUENCES, 865 ONSET EVENTS). θ^* MAXIMISES $J = q_{in} - q_{out}$ AT $w = 5$ SUBJECT TO $a_G(R = 30) + c \le 0.25$; J_{free}: MAXIMUM WITHOUT THE CONSTRAINT; c: CUE COST IN DETECTOR EQUIVALENTS (SAME BENCHMARK SESSION); q_b: BACKGROUND FIRING PER EGO CLASS. PRE-REGISTERED (TAG `PREREG-28-V1`).

cue	θ^*	q_{in}	q_{out}	J	J_{free}	a_G	c	q_b hov./mov.
border band	0.0768	0.101	0.065	0.036	0.078	0.127	0.026	0.031/0.148
ORB-lite diff.	0.0436	0.144	0.093	0.052	0.134	0.228	0.098	0.061/0.170
raw diff.	0.0279	0.249	0.192	0.058	0.118	0.249	0.023	0.129/0.346
tiny det.	0.8833	0.111	0.063	0.048	0.077	0.227	0.126	0.021/0.000
ORB diff. (full)		excluded: $c \ge 1$ (Thm. 2)					1.024	–

at reduced width), *border band* (the same, restricted to a band along the image border, aimed at vehicles entering the footprint), *ORB-lite difference* (difference after a light homography compensation) and *tiny detector* (maximum confidence of YOLO26-n at input 320). A fifth cue with full ORB compensation was excluded by a pre-registered cost rule: its cost exceeded one detector run ($c \ge 1$), which by Theorem 2 rules out any gain. The cue cost c is the ratio of the median cue time to the median detector time measured in the same benchmark session on 200 TRAIN frame pairs (single-thread OpenCV cues, tiny detector on the CPU, main detector on the integrated GPU; detector 94.6 ms/frame). Across sessions the denominator ranged from 62.9 to 94.6 ms/frame; c uses only the session that also timed the cues.

Each cue gets one threshold θ^*, chosen on TRAIN by a rule fixed before any cue was traced: maximise the onset separation $J = q_{in} - q_{out}$ at onset window $w = 5$, subject to $a_G(R = 30) + c \le 0.25$. Here q_{in} is the firing rate on frames inside some onset window and q_{out} outside all windows; q_b and q_o of Lemma 2 are fitted by maximum likelihood on the out-of-window frames, per ego class. Table I lists the operating points. All cues are weak ($J \le 0.058$), and the cost constraint is binding for every cue.

E. Policies, KPIs and Comparisons

The gate runs the detector on a frame if the cue fires (score $\ge \theta^*$) or if the frame is a refresh frame ($R = 30$, uniform phase); its cost is $a_G + c$, with the cue computed on every frame. The periodic competitor uses a real-valued stride with uniformly distributed phase. Both are replayed offline on the same detector dump and cue traces. The KPI is the onset latency: an event is missed at level L_{max} if no successful detector run falls on its first $\min(D_e, L_{max} + 1)$ frames, and $L_{max} \in \{3, 5\}$ frames (0.10 and 0.17 s). Three comparisons were pre-registered:

- *matched cost* (H3/H5)[1]: $\hat{\Delta} = \overline{\text{miss}}_P - \overline{\text{miss}}_G$ with the periodic stride set so that $a_P = a_G + c$, matched within each ego stratum;
- *iso-KPI* (H2): the gate is cheaper if it reaches latency-miss $\le \varepsilon = 2\%$ and $a_G + c < a_P(\varepsilon)$, where $a_P(\varepsilon)$ is the activation of the largest periodic stride that reaches ε;
- *recall pair* (H8): $\widehat{\Delta\Delta} = \hat{\Delta}(r_{low}) - \hat{\Delta}(r_{main})$ on the same events and the same gate schedule; Remark 1 predicts an increase when both recalls exceed $r^\dagger$ of the configuration (largest $r^\dagger$ on TRAIN: 0.723).

[1]Hypothesis numbers follow the pre-registration; H1, H4, H6 and H7 are descriptive and were not scored.

F. Predictions

For each cell the predicted Δ uses Lemma 1 for the periodic policy and Proposition 2 for the gate, averaged over the TRAIN durations of the cell's events. Two departures from the idealised theory were fixed before freezing: frames of the KPI window after the event's own onset window fire at the mean-field rate $\rho q_{in} + (1 - \rho) q_{out}$ instead of q_{out} (A-sparse is too conservative for long windows), and q_{out} is replaced by $q_{out,eff}(M)$ of Lemma 2 at a representative density of the cell's bin. The predicted sign is "+" if $\Delta_{pred} > \delta_{min} = 0.005$ and "≤ 0" otherwise; for H2 it is "+" if the predicted gate reaches ε and is cheaper.

G. Confirmatory Family

The full grid (four cues, two KPIs, iso-KPI targets, density bins, ego classes, recall pair) gave 144 cells that pass the size rule of at least 30 events in at least 3 sequences on TRAIN. A reduction order fixed in advance and depending only on cell counts brought it below the limit of 80 cells: dropping $\varepsilon = 5\%$ left 108 and dropping $L_{max} = 10$ left 72. The family has three strata: hovering M 6–20 (215 TRAIN events in 15 sequences), hovering $M > 20$ (310 / 12) and moving M 6–20 (167 / 10); the other strata are too small. In each stratum and for each cue there are two H2, two H3/H5 and two H8 cells ($L_{max} \in \{3, 5\}$). The model predicted a gate advantage in 21 cells: 2 matched-cost cells (border band, hovering, $L_{max} = 3$, both densities) and 19 H8 cells.

H. Scoring, Criteria and Statistics

Confidence intervals are 95% paired bootstrap intervals over sequences ($B = 1{,}000$, seed 42). A prediction "+" is *correct* if the lower limit of $\hat{\Delta}$ is above zero, *wrong* if the upper limit is at most zero, and *inconclusive* otherwise; a prediction "≤ 0" is correct if the upper limit is below δ_{min}, wrong if the lower limit is above δ_{min}, and inconclusive otherwise. H8 uses the same rule on $\widehat{\Delta\Delta}$. An H2 cell is correct or wrong according to whether "gate cheaper at equal KPI" matches the prediction, and inconclusive if the interval of $a_G + c - a_P(\varepsilon)$ contains zero. The verdict rules were fixed in this order:

P *Power*: if more than 50% of the cells are inconclusive, the conclusion is "not testable with these data".
(i) Refuted if fewer than 80% of the decided cells are correct.
(ii) Refuted if the observed iso-KPI boundary q_o^* lies outside its predicted interval in at least two thirds of the density groups.
(iii) Refuted if the H8 sign is wrong in at least 50% of the decided H8 cells.

Holm's correction within the family is reported only as a count; it does not define correctness. The test split was opened once, on 3 October 2026, with the frozen code; the implementation choices that the pre-registration left open were recorded before any test-split effect was computed (Appendix B).

TABLE II
CONFIRMATORY FAMILY ON UAVDT TEST (PRE-REGISTERED, 72 CELLS, THREE-WAY SCORING). LAST ROW (POST HOC): THE 24 ISO-KPI CELLS (H2) REMOVED BECAUSE NEITHER POLICY CAN REACH $\varepsilon = 2\%$ (SEC. VI-F).

group	cells	correct	wrong	inconcl.	correct/decided
H2 (iso-KPI)	24	24	0	0	–
H3/H5 (matched cost)	24	21	1	2	–
H8 (recall pair)	24	17	4	3	–
all	72	62	5	5	92.5%
without H2 (post hoc)	48	38	5	5	88.4%

VI. RESULTS

A. The Test Split

The 20 TEST sequences contain 839 onset events (364 entering from the border, 475 in the interior), of which 737 lie in the birth window. With the frozen ego threshold, 13 sequences are hovering and 7 moving. The three confirmatory strata pass the size rule on TEST: hovering M 6–20 with 121 events in 8 sequences, hovering $M > 20$ with 332 in 9, and moving M 6–20 with 221 in 7. The main and the low-recall detector ran at 99.2 and 30.4 ms/frame on the integrated GPU (median over sequences).

B. Confirmatory Family

Table II and Fig. 2 summarise the 72 cells. 62 are correct, 5 wrong and 5 inconclusive. In the order fixed in advance:

- P 6.9% of the cells are inconclusive, below 50%: the family has enough power to be scored.
- (i) 92.5% of the 67 decided cells are correct, not below 80%: not refuted.
- (ii) *Not evaluated.* The criterion compares the observed iso-KPI boundary q_o^* with its predicted interval, but the frozen document contains a predicted interval only for the simulated channel used to test the pipeline, not for the real TRAIN channel. The criterion therefore cannot be applied, and we do not substitute an interval computed after unblinding; the predicted boundary and a post hoc observed one are shown descriptively in Supplementary S2 and S4.
- (iii) 19.0% of the decided H8 cells are wrong, below 50%: not refuted.

By the rule fixed in advance the result is therefore "confirmed", resting on criteria (i) and (iii) only. Holm's correction leaves 22 of the cells significant.

The share of correct cells overstates the evidence. All 24 iso-KPI cells (H2) predicted "gate not cheaper" and all are correct, but for a reason the pre-registration did not anticipate: on TEST no periodic stride reaches latency-miss $\varepsilon = 2\%$, not even running the detector on every frame, so $a_P(\varepsilon) = \infty$, and the gate does not reach it either (Section VI-F). These cells do not discriminate between the theory and its negation. Removing them, 38 of the 43 decided cells are correct (88.4%).

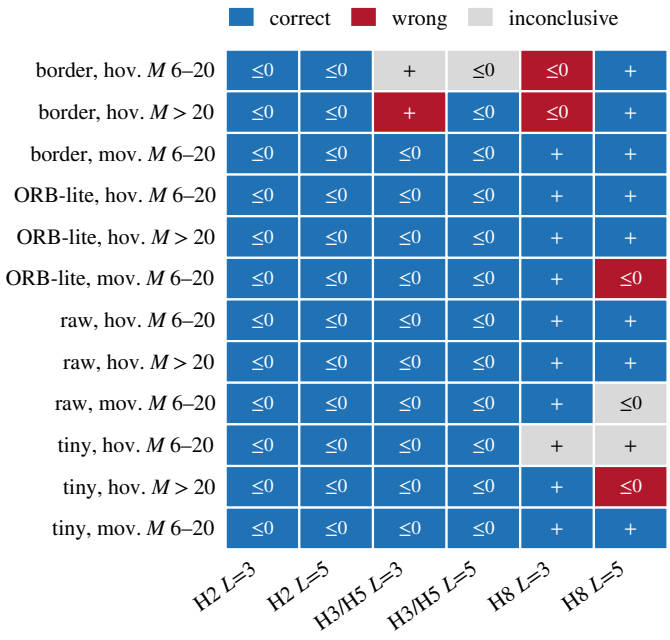


Fig. 2. Confirmatory family on UAVDT TEST (pre-registered). Rows: cue, ego class and density bin; columns: hypothesis and latency target L_{max}. Text: the frozen prediction ("+" = gate advantage); colour: three-way result.

C. The Predicted Gate Corner

The model predicted a matched-cost advantage for the gate in exactly the corner suggested by the theory: the border-band cue, hovering camera, $L_{max} = 3$, in both density bins (Supplementary Table S5). The corner did not appear. In hovering $M > 20$ the gate lost, $\hat{\Delta} = -0.134$ [-0.193, -0.098], against a prediction of +0.040; this is the only wrong matched-cost cell. In hovering M 6–20 the point estimate is also negative, -0.099 [-0.210, +0.220], and the cell is inconclusive. Across all 24 matched-cost cells the largest point estimate is -0.099 and 0 cells have a confidence interval above zero: the periodic schedule was never beaten. The 22 "≤ 0" predictions were correct in 21 cells and inconclusive in 1 (Fig. 3, left).

D. Recall Pair (H8)

Replacing the main detector by the lower-recall one increased $\hat{\Delta}$ in all 24 H8 cells (smallest $\widehat{\Delta\Delta} = +0.024$), with an interval above zero in 22 (Fig. 3, right). This is the direction of Remark 1: a worse detector gives the gate's consecutive looks more value. Of the 19 cells with a predicted increase, 17 are correct and 2 inconclusive. The 4 wrong H8 cells are all cells in which the model predicted no increase beyond δ_{min} and the increase occurred anyway. The effect is real but it does not rescue the gate: even with the lower-recall detector the gate loses at matched cost, because $\hat{\Delta}$ starts from a negative value.

E. Descriptive Variants

Supplementary Table S7 repeats the scoring of the same 72 cells under four variants that were declared descriptive in advance. Moving the onset to the earliest frame in which

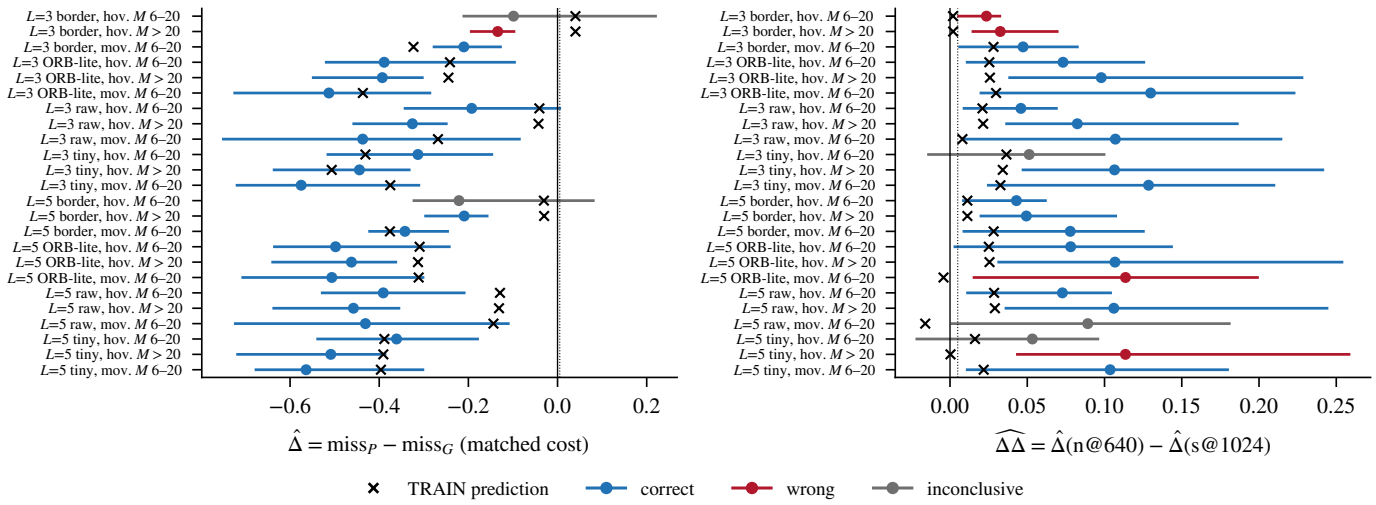


Fig. 3. UAVDT TEST, pre-registered. Left: matched-cost effect $\hat{\Delta}$ (positive: the gate misses fewer onsets at equal cost) for the 24 H3/H5 cells. Right: recall-pair effect $\widehat{\Delta\Delta}$ for the 24 H8 cells. Bars: 95% paired sequence-bootstrap CI ($B = 1{,}000$); crosses: frozen TRAIN predictions; dotted line: $\delta_{min} = 0.005$.

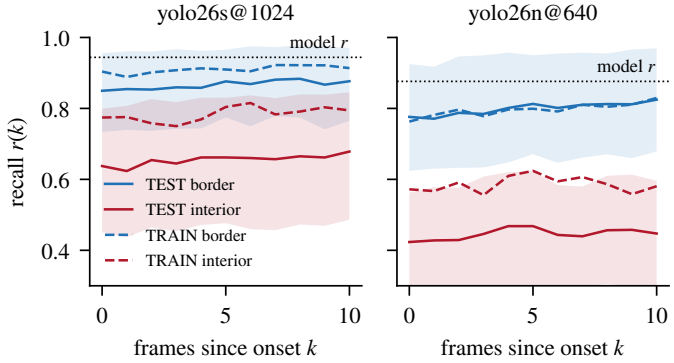


Fig. 4. Post hoc. Recall $r(k)$ on frame k after onset (score ≥ 0.25, IoU ≥ 0.5) for onset events entering from the border and appearing in the interior, TEST (solid, band: 95% sequence-bootstrap CI) and TRAIN (dashed). Dotted: the constant recall used by the frozen model.

the detector already saw the vehicle (onset$_{alt}$, motivated by the audit below) and lowering the detector score threshold to 0.05 increase the number of inconclusive cells; the region-of-interest variant and the higher score threshold leave the picture unchanged. None of the variants produces a matched-cost cell in which the gate wins.

F. Why the Corner Did Not Appear (Post Hoc)

The analyses in this subsection were designed after the test split was opened. They are not part of the confirmatory family and do not change its verdict.

Weak cues. At the frozen thresholds every cue separates onset frames from other frames only weakly, $J \leq 0.058$ on TRAIN (Table I), and the cost constraint $a_G \leq 0.25$ binds for every cue, so the separation available without the constraint (J_{free}) cannot be used. On TEST the separation is smaller again, at most 0.020 (Supplementary Table S4).

Recall at onset. The model uses the average per-look recall $r = 0.944$, but for a vehicle that has just appeared the recall is much lower (Fig. 4): on TEST, $r(k)$ for the first frames after onset is 0.731 for all events, 0.850 for border entries and 0.637 for interior onsets (median box size 15.5 px), against 0.858 on TRAIN. Every schedule therefore has a latency-miss floor (Supplementary Table S6): running the detector on every frame still misses between 4.1% and 32.2% of onsets, depending on the stratum and on L_{max}, and the best gate on the TRAIN threshold grid misses between 11.6% and 59.6%. The floor is far above what independent misses at $r(0)$ would give, which indicates that misses at onset are persistent; we did not measure the autocorrelation directly. The frozen target $\varepsilon = 2\%$ lies below every floor, which is why the H2 cells are empty, and even at relaxed targets (the floor plus two percentage points, or fixed levels up to twenty percent) the gate at its frozen threshold does not reach the target in any stratum (Supplementary Fig. S2).

Calibration of the model. The frozen mean-field model ranks the cells correctly but is optimistic about the gate (Table III, Supplementary Fig. S3): over the 48 matched-cost and H8 cells the Spearman correlation between predicted and observed values is 0.785, while the observed matched-cost $\hat{\Delta}$ is below the prediction in 19 of 24 cells, by a median of -0.160 (border band, hovering M 6–20, $L_{max} = 5$: predicted -0.031, observed -0.221 [-0.322, +0.080]). For H8 the model under-predicts the increase (bias +0.062) and the rank correlation is weak (0.198). Both deviations follow from the onset-recall measurements: a constant, independent per-look recall over-credits the gate's consecutive looks at matched cost and under-credits the cost of a worse detector.

Channel drift. The TEST sequences are busier and the cues separate onsets less well than on TRAIN: the onset rate rises from 0.0363 to 0.0506 per frame, and J at the frozen thresholds falls by a factor of 2.9 to 6.3 across the four cues (Supplementary Table S4). The wrong border-band cell follows from this drift: in hovering $M > 20$ the border band fired inside onset windows with probability 0.1087 on TRAIN but only 0.0293 on TEST, below its out-of-window rate 0.0359, so its gate missed 0.893 of onsets at $L_{max} = 3$

TABLE III
CALIBRATION OF THE FROZEN MEAN-FIELD PREDICTIONS AGAINST UAVDT TEST (POST HOC): ORDINARY LEAST SQUARES OF THE OBSERVED ON THE PREDICTED VALUE, MEAN ABSOLUTE ERROR, MEAN BIAS (OBSERVED − PREDICTED) AND SPEARMAN RANK CORRELATION.

cells	n	slope	intercept	MAE	bias	Spearman
H3/H5 (Δ)	24	0.567	-0.238	0.163	-0.133	0.623
H8 ($\Delta\Delta$)	24	0.363	+0.074	0.062	+0.062	0.198
both	48	1.230	-0.010	0.112	-0.036	0.785

against a predicted 0.590. In hovering M 6–20 the separation improved on TEST (+0.0754 against +0.0103), but the gate's miss (0.748 against 0.594 predicted) was raised by the low onset recall, and with 8 sequences the interval is wide.

VII. DISCUSSION AND DESIGN RULE

A. A Design Rule for the Detector Rate

The theory and the experiment together suggest the following procedure for an onboard UAV traffic pipeline with current cheap cues and a modern detector. It is a consequence of the results above for this setting, not a universal prescription.

(a) Measure the floor. Run the detector on every frame over representative footage and measure the latency-miss of newly appearing vehicles at the target L_{max}. No schedule can do better (Supplementary Table S6); on UAVDT TEST the floor ranged from 4.1% to 32.2%. A target below the floor calls for a better detector, not a better schedule.

(b) Choose the periodic stride. Above the floor, choose the largest stride $S^*(\varepsilon)$ that meets the target, from Lemma 1 applied to the latency window and the measured duration law. As an illustration only (ideal detector, $r = 1$, miss KPI, UAVDT TRAIN events), the miss KPI needs $S^* = 20$ (1.5 detector runs per second) at $\varepsilon = 5\%$ and $S^* = 6$ (5.0 Hz) at $\varepsilon = 2\%$; with a real detector the measured floor of step (a) comes first. When the footage is not representative, the kinematic certificate of Theorem 1 gives a phase-free requirement, $f_{det} \geq n(r, \varepsilon)\, v_{max}/L(h)$: for a nadir camera at 100 m with a 60° field of view, $v_{max} = 25$ m/s and $\varepsilon = 1\%$ (illustrative values, not UAVDT metadata), it asks for 0.650 Hz at $r = 0.8$. The recall to insert is the recall at onset for the target class, not the average recall (Fig. 4), and because the measured floors indicate that misses at onset are persistent, the certificate should be validated against the measured floor.

(c) Screen any gate. A gate is worth replaying only if its measured onset separation exceeds the threshold of Corollary 3, $J > J^* = Tc/(w' - T\rho_w)$. The threshold depends only on the cue cost c, the onset-window density ρ_w and the KPI window, not on the detector's recall. For the border band at $L_{max} = 3$, $J^* = 0.030$ with the TRAIN onset density: the TRAIN separation 0.036 passed the screen, consistent with the model's predicted narrow win there, while the TEST separation 0.011 did not. The tiny detector and the ORB-lite cue fail it even on TRAIN ($J^* = 0.147$ and 0.114) because they are expensive relative to their separation. Passing the screen is necessary, not sufficient; the full matched-cost or iso-KPI comparison must follow, ideally with the onset recall measured on the deployment data.

B. Where Gating Remains Useful

The negative result concerns frame-level gates driven by cheap image cues. It does not exclude gates of other kinds. Theorem 3 locates the obstacle in the false-fire rate $q_{out,eff}$, which grows with the number of other vehicles and with ego-motion; a gate that reasons about regions, for example a cue restricted to the entry zones of a road or to areas without tracked vehicles [23], attacks that term directly. A cue with a much larger onset separation, such as a learned onset detector, could pass the screen of Corollary 3. And loose latency targets or sparse rural traffic move the operating point towards the corner in which the theory leaves room. In the regime tested here, busy urban scenes from a hovering or moving drone with a latency target of a fraction of a second, a periodic schedule at the stride of step (b) is the simpler and the safer design.

C. Limitations

Empty iso-KPI cells. The frozen target $\varepsilon = 2\%$ lies below the latency-miss floor of every schedule on TEST. The model predicted a finite $a_P(\varepsilon)$ because it assumes a constant, independent per-look recall; on the data $a_P(\varepsilon) = \infty$. The 24 H2 cells are therefore correct without testing the iso-KPI boundary, and we report the share of correct cells with and without them.

Criterion (ii). The density-boundary criterion could not be evaluated because no predicted interval for q_o^* was frozen for the real channel. This was an omission in the pre-registration; the post hoc description in Section VI-F cannot replace it.

Annotation timing. The audit of the five frames before onset, fixed in advance as descriptive, found that the detector already saw 68.4% of interior onsets on TEST (67.7% on TRAIN) at least three times before the annotation starts, without any earlier annotation of the track. Onset times of interior events are thus partly annotation artefacts. Moving the onset earlier (onset_{alt}) changed no predicted sign on TRAIN (largest change of Δ_{pred}: 0.0096), and on TEST it changed the scoring only towards more inconclusive cells (Supplementary Table S7).

Sample size. Each confirmatory stratum contains between 7 and 9 TEST sequences. Sequence-level bootstrap intervals are correspondingly wide, which explains most inconclusive cells.

One dataset, one platform. All results come from UAVDT; the VisDrone-VID replication listed in the pre-registration is left for a follow-up study. Cue costs were measured on one laptop with an integrated GPU, with the cue and the detector timed in the same session; the detector time varied between 62.9 and 94.6 ms/frame across sessions, so the cost ratio c, and through it every θ^*, is platform-specific.

Pre-registration history. A version of the pre-registration that removed events judged "annotated late" from the event set was withdrawn before the freeze because its audit rule was wrong; no number from it is used. Appendix A lists all versions.

VIII. CONCLUSION

We asked when an oblivious periodic detector schedule is beaten, at equal cost, by a cue-triggered gate for newly

appearing vehicles in UAV traffic video. The theory gives exact miss and latency expressions for both policies and shows that a frame-level gate can win only in a narrow corner, which closes as traffic density, ego-motion and cue cost grow. In a pre-registered test on UAVDT the frozen criteria are not violated (62 correct, 5 wrong and 5 inconclusive cells out of 72), with two qualifications: the iso-KPI cells were uninformative because no schedule could reach the frozen target, and the density-boundary criterion could not be evaluated. The corner in which the model predicted a gate win did not appear, and no gate beat the periodic schedule in any matched-cost cell. Post hoc measurements locate the reasons in weak cues, low detector recall at onset and drift of the cue channel, and show that the model ranks the cells well but is optimistic about the gate. For practice: measure the latency-miss floor, choose the periodic stride from the iso-miss requirement or the kinematic certificate, and replay a gate only if its onset separation exceeds the threshold implied by the gain–cost identity.

Reproducibility and Data

Code, frozen JSON files, per-cell results (all 72 cells: Supplementary Tables S1–S3) and the generator of every number in this paper are at https://github.com/lamquocdat01/uav-periodic-vs-gating (release v1.0-submission). The pre-registration (`PREREG_28.md`, sha256 `81c19b43473e54d9be13be4e7ecaebbaab9e35d479f6027b247afc6b0a3640ab`) was frozen with tag `prereg-28-v1` (commit `5cfe901`, 1 October 2026) together with the hashes of the TRAIN channel, the cell list (sha256 prefix `fb30790c2057755d`), the code and the detector weights; the test split was opened once, on 3 October 2026, with the frozen scorer, and every deviation is logged. Bootstrap: $B = 1{,}000$, seed 42; Python 3.11.9, Ultralytics 8.4.163, OpenVINO 2026.4. UAVDT [16] and VisDrone [17] are available from their official sources; frames are not redistributed.

Appendix A
Pre-Registration History

The rules were fixed in steps, each recorded in the project log with the reason for the change; the TEST split was not read at any step.

- 27 September: scoring rule and refutation criteria (i)–(iii); confirmatory family restricted to latency KPIs, iso-KPI and the recall pair.
- 28 September: one threshold per cue by the constrained Youden rule; family reduction order depending only on cell counts. Both rules were fixed before any real cue trace existed and tested on a simulated channel.
- 30 September: detector weights, cue costs from a same-session benchmark, recall of four detector levels on TRAIN.
- 1 October (morning): detection score threshold 0.25; rule for the H8 detector pair; real image sizes; and an audit rule that removed from the onset events every event the detector had seen in at least three of the five frames before its first annotation. This rule removed 82% of the TRAIN events (153 kept). It was *withdrawn* the same day before the freeze, because it classified vehicles that enter the image partly visible, and tracks that are annotated but occluded before onset, as late annotations; no number of that version is used anywhere.
- 1 October (afternoon): the audit kept as a descriptive analysis only (no event removed), with five classes (partial entry, visibility transition, late annotation, true birth, censored start); all onset events kept; channel, thresholds and family rebuilt on TRAIN.
- 1 October (evening): three-way scoring with the margin δ_{min}, the power criterion P, and the $r^{\dagger}$ condition for H8 cells (no H8 cell was removed by it).
- 1 October 2026, 17:04 (UTC+7): freeze (tag `prereg-28-v1`, commit `5cfe901`) of the pre-registration, channel, cells, code and weights hashes.

Appendix B
Implementation Choices Fixed Before Computing Test Effects

Choices the frozen documents left open, recorded in the replay script and the log before any TEST effect was computed:

(a) a detection is a dump box with score ≥ 0.25, as for the TRAIN recall;
(b) the periodic stride is matched to the gate's cost within the timeline of each ego stratum (ego class is a sequence attribute and the predictions use stratum-specific costs; density bins select events within that timeline);
(c) the H8 interval is the paired bootstrap interval of the per-event difference of the two detectors' Δ;
(d) for H2 the point value of $a_P(\varepsilon)$ uses the frozen replay engine and its interval a geometric grid of strides;
(e) a cell failing the size rule on TEST would be scored inconclusive and kept (none failed); the cell construction reproduces the frozen TRAIN event and sequence counts of every cell, and TEST ego-motion uses the frozen procedure and threshold;
(f) criterion (ii) is not evaluated, because no predicted interval exists for the real channel.

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